



TESTING SERVICES

LABORATORY TEST REPORT

Report # 83106B
Lab Test Number: 3268-6821
Report Date: August 23, 2021

ASTM C67, D573, F1292 Freeze-Thaw Cycling

www.testingservices-usa.com ▪ (706)226-1400

office@testingservices-usa.com

CLIENT:

Company:	Pro-Techs Surfacing, LLC
Address:	2072 Coon Road
	Akron, OH 44321
Requested By:	Tracy Slegus

TEST MATERIAL:

Date Material Received:	May 31, 2021
Material Type:	Poured-in-place Safety Surface
Material Condition:	Excellent, New
Sample ID:	Perma Play 2 layer system; 3.5" Overall Thickness ½" EPDM Top Layer + 3.0" SBR base
Subbase:	Concrete

TESTING METHODS REQUESTED:

Testing Services Inc. was instructed by the client to test for the following...			
Standard:	ASTM C67/C67M	Test Method:	Standard Test Methods for Sampling and Testing Brick and Structural Clay (modified)
Standard:	ASTM D573	Test Method:	Standard Test Method for Rubber – Deterioration in an Air Oven
Standard:	ASTM F1292	Test Method:	Standard Specification for Impact Attenuation of Surfacing Materials Within the Use Zone of Playground Equipment

SAMPLING PLAN:

Sampling Date:	6/2/2021
- Specimen sampling is performed in the sampling department at TSI. - The sampling size of specimens is determined by the test method requirements. - In the event a specific sampling size is not called for, a determination will be made based on previous testing experience, and approved for use by an authorized manager. - All samples are subjected to the outside environmental conditions of temperature and relative humidity. - Sample requiring pre-determined exposure to specified environmental conditions based on a specific test method, take place in the departments in which they are tested	

DEVIATION FROM TEST METHOD:

State reason for any Deviation from, Additions to, or Exclusions From Test Method.
None

TEST SCOPE:

These test method covers the procedure to deterioration and critical fall height of submitted rubber interlocking tiles after resistance to Freeze-thaw cycling and exposure to heat.

TEST PROCEDURE:

A 12" x 12" section of the test material was selected and placed in standard lab conditions, 70°F 65% RH, for 48 hours prior to testing. Initial prep for Freeze/thaw was performed by submerging the specimen in a thawing tank of water (75 ± 10°F) for 4 ± ½ hours. Freeze/thaw cycling then began. One cycle consists of placing specimen in tray of ½ inch water, which is put in freezing chamber for 20 ± 1 hour (not to exceed 16°F). The specimen is then submerged in the thawing tank for 4 ± ½ hours. This procedure was followed for a total of 50 cycles. After the 50 cycles of freeze/thaw exposure, the specimen was placed in an air circulating oven set at 200°F for a duration of 7 days. The specimen was tested for critical fall height at ambient temperatures.

TEST SUMMARY:

TEST METHOD	TEST DESCRIPTION	TEST RESULT	
ASTM C67/C67M-21	Freezing and Thawing; 50 cycles	Deterioration not present	
ASTM D573-04(2019)	Deterioration in an Air Oven	Deterioration not present	
ASTM F1292-18e1	Impact Attenuation; Ambient Temp only, Concrete subbase	8' CFH	Gmax 148 / HIC 968

Uncertainty:

We undertake all assignments for our clients on a best effort basis. Our findings and judgments are based on the information using the latest test methods available. TSI can only ensure the test results for the specific items tested. Unless otherwise noted in the deviations sections of this report, all tests performed are in compliance with stated test method.

Test Report Approval:

 Digitally signed by Erle Miles, III, Lab Director, Testing Services (TSI) LLC
Erle Miles, III, Lab Director, Testing Services (TSI) LLC

Testing Services (TSI) LLC
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03/14/2023

To whom it may concern:

All products within the current **Berliner Seilfabrik** catalog as well as the **Urban Design Berlin** catalog, produced by Berliner Seilfabrik GmbH & Co. and exclusively distributed in the U.S. by Berliner Seilfabrik Play Equipment Corporation and their dealer network, are **compliant to the ASTM F1487 and the CPSC guidelines**. Following the CPSC guidelines, flexible climbers are not intended for the use by the age-group 2-5. If installed in conjunction with an ADA compliant surfacing, climbers also meet the ADA requirements.

Best regards,



David Koehler

(IPEMA certified Inspector &
President of Berliner Seilfabrik Play Equipment Corporation)

**TESTING SERVICES, INC.**

817 SHOWALTER AVE., PO BOX 2041

DALTON, GA 30722-2041

PHONE: (706)226-1400 FAX: (706)226-6118



CLIENT:	ForeverLawn	REPORT NUMBER:	61865
	5801 Mayfair Rd, Suite 4	LAB TEST NUMBER:	2650-1188
	North Canton, OH 44720	DATE:	September 17, 2014

SAMPLE ID:

Turf with Infill	FLS48 Playground Academy with infill installed of 0.75 lbs/ft ² 100% 10/20 SBR
Layer 1: (Under Turf)	3" Foam Pad
Layer 2: (Under Pad)	2" Aggregate Sub-Base

INTRODUCTION:

Testing Services Inc was instructed by the client, to perform ADA wheelchair accessibility for the above described material being used under and around playground equipment.

TEST METHOD:

ASTM F1951: Standard Specification for Determination of Accessibility of Surface Systems Under and Around Playground Equipment

REQUIREMENT:

A surface in place shall have average work per foot (work per meter) values for straight propulsion and for turning **less** than the average work per foot (work per meter) values for straight propulsion and for turning, respectively, on a hard, smooth, surface with a grade of 1:14 (7.1 %).

PROCEDURE:

Test Surface Preparation: Tests were conducted on 09/17/14 indoors at TSi Laboratories in an environment of 78°F and 44% R.H. The synthetic turf was installed over the above identified layers and infilled with the above identified infill in a wooden box (44"W x 117"L). The system, prior to testing, was slightly compacted using a water-filled lawn roller to simulate foot traffic.

Wheelchair/Operator: The wheelchair used in these tests was manufactured by *Invcare*, Model Action Xtra, serial Number 98J84142. This wheelchair is totally adjustable, a necessity for these tests. The pneumatic tires were inflated to 60 psi on the rear and 32 psi on the front. The weight of the wheelchair was 24.25 pounds and the operator's weight was 165 pounds for a total of 189 pounds. The operator's distribution was adjusted to 60% on the rear wheels and 40 % on the front.

Torque Measuring System: A certified *Dillon Electronic Force Gauge*, Model BFG 500N, S/N 98-2277-07 was used as an interface between a *Dell* Laptop and a certified *Dillon Smart Torque Wrench*, S/N 97-0085-01. Software, also from Dillon, logged the load vs. time and integrated the area under the resulting curves. The adapters and accessories needed to attach the instrumentation were fabricated locally. This total package added 10 pounds to the total weight bringing the total to 199 pounds.

TEST RESULTS: (Average Work/ft-Force)

Baseline Straight	FLS48 Playground Academy turf with above listed infill over above listed
14.33 lbs	8.82 lbs
Baseline Turning	FLS48 Playground Academy turf with above listed infill over above listed
10.14 lbs	7.32 lbs

CONCLUSION:

The above listed material *meets/exceeds* both the straight line and turning propulsion requirements set forth in this test method and therefore, passes the standard.


Digitally signed by Erle Miles, Jr.
VP
DN: cn=Erle Miles, Jr. VP,
o=Testing Services Inc., ou,
email=tsi@windstream.net,
c=US
Date: 2014.09.18 13:35:59 -0400

Erle Miles, Jr V.P., Testing Services Inc

TSi Accreditation:

Our laboratory is accredited with US Dept of Commerce, National Institute of Standards and Technology: ISO/IEC 17025:2005. Our code # is NVLAP 100108-0. However, it should be noted that some or all of the tests performed are not under our scope of accreditation due to the work not fully conforming to the standard, or it being outside the scope of our accreditation, or subcontracted.

OUR LETTERS AND REPORTS APPLY ONLY TO THE SAMPLE TESTED AND ARE NOT NECESSARILY INDICATIVE OF THE QUALITIES OF APPARENTLY IDENTICAL OR SIMILAR PRODUCTS. THESE LETTERS AND REPORTS ARE FOR THE USE ONLY OF THE CLIENT TO WHOM THEY ARE ADDRESSED AND THEIR COMMUNICATION TO ANY OTHERS OR THE USE OF THE NAME TESTING SERVICES, INC. MUST RECEIVE OUR PRIOR WRITTEN APPROVAL. THE REPORTS AND LETTERS, AND OUR NAME, OUR SEALS, OR OUR INSIGNIA ARE NOT UNDER ANY CIRCUMSTANCES TO BE USED IN ADVERTISING TO THE GENERAL PUBLIC.



CLIENT:

Company:	Pro-Techs Surfacing, LLC
Address:	2072 Coon Road
	Akron, OH 44321
Requested By:	Tracy Slegus

TEST MATERIAL:

Date Material Received:	May 31, 2021
Material Type:	Poured-in-place Safety Surface
Material Condition:	Excellent, New
Sample ID:	Perma Play 2 layer system; 5" Overall Thickness ½" EPDM Top Layer + 4.5" SBR base
Subbase:	Concrete

TESTING METHODS REQUESTED:

Testing Services, Inc was instructed by the client to perform the following testing..			
Standard:	ASTM F1951-14	Test Method:	Standard Specification for Determination of Accessibility of Surface Systems Under and Around Playground Equipment

SAMPLING PLAN:

Sampling Date:	6/2/2021
- Specimen sampling is performed in the sampling department at TSI. - The sampling size of specimens is determined by the test method requirements. - In the event a specific sampling size is not called for, a determination will be made based on previous testing experience, and approved for use by an authorized manager. - All samples are subjected to the outside environmental conditions of temperature and relative humidity. - Sample requiring pre-determined exposure to specified environmental conditions based on a specific test method, take place in the departments in which they are tested	

DEVIATION FROM TEST METHOD:

State reason for any Deviation from, Additions to, or Exclusions From Test Method.
none

REQUIREMENT:

A surface in place shall have average work per foot (work per meter) values for straight propulsion and for turning **less** than the average work per foot (work per meter) values for straight propulsion and for turning, respectively, on a hard, smooth, surface with a grade of 1:14 (7.1 %).

PROCEDURE:

Test Surface Preparation: Tests were conducted on 6/3/2021 indoors at TSi Laboratories in an environment of 74°F and 60% R.H. The pour-in-place system was installed over above listed sub-base.

Wheelchair/Operator: The wheelchair used in these tests was manufactured by *Invcare*, Model Action Xtra, Serial Number 98J84142. This wheelchair is totally adjustable, a necessity for these tests. The pneumatic tires were inflated to 60 psi on the rear and 32 psi on the front. The weight of the wheelchair was 24.25 pounds and the operator's weight was 165 pounds for a total of 189 pounds. The operator's distribution was adjusted to 60% on the rear wheels and 40 % on the front.

Torque Measuring System: A certified *Mecmesin Advanced Force Gauge*, Model 500N, was used as an interface between a *Dell* Laptop and a calibrated *Smart Torque Wrench*, S/N 97-0085-01. Emperor Lite Software, from *Mecmesin*, logged the load vs. time and integrated the area under the resulting curves. The adapters and accessories needed to attach the instrumentation were fabricated locally. This total package added 10 pounds to the total weight bringing the total to 199 pounds.



CLIENT:

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Address:	2072 Coon Road
	Akron, OH 44321
Requested By:	Tracy Slegus

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Sample ID:	Perma Play 2 layer system; 5" Overall Thickness ½" EPDM Top Layer + 4.5" SBR base
Subbase:	Concrete

TEST SUMMARY:

TEST METHOD	Maximum Requirements – Average Work/ft-Force	TEST RESULTS – Average Work/ft-Force
ASTM F1951-14	Baseline Straight: 15.08 lbs	5.53 lbs
	Baseline Turning: 11.05 lbs	7.23 lbs

Straight Propulsion	1	2	3	4	5
Circumference of Rear Wheel	75.375"	75.375"	75.375"	75.375"	75.375"
Area	23.4977 ft ² lbs*s	22.1574 ft ² lbs*s	21.1461 ft ² lbs*s	18.5261 ft ² lbs*s	20.9106 ft ² lbs*s
Time	7.80 seconds	7.77 seconds	7.75 seconds	7.69 seconds	7.73 seconds
Distance	86.0 inches	86.0 inches	86.0 inches	86.0 inches	86.0 inches
Distance	7.17 ft	7.17 ft	7.17 ft	7.17 ft	7.17 ft
Angular Displacement (radians)	7.17 rad	7.17 rad	7.17 rad	7.17 rad	7.17 rad
Average Torque (energy)	3.01 ft lbs	2.85 ft lbs	2.73 ft lbs	2.41 ft lbs	2.71 ft lbs
Total Work (energy)	43.19 ft lbs	40.89 ft lbs	39.12 ft lbs	34.54 ft lbs	38.79 ft lbs
Work/ft (force)	6.03 lbs	5.71 lbs	5.46 lbs	4.82 lbs	5.41 lbs
Drop Hi/Low Work/ft (force)		5.71 lbs	5.46 lbs		5.41 lbs
Average Work/ft (force)			5.53 lbs		

Turning Propulsion	1	2	3	4	5
Circumference of Rear Wheel	75.375"	75.375"	75.375"	75.375"	75.375"
Distance from Pivot Point to Outer Wheel	35.75 inches	35.75 inches	35.75 inches	35.75 inches	35.75 inches
Area	54.2215 ft ² lbs*s	55.8756 ft ² lbs*s	53.9420 ft ² lbs*s	58.3214 ft ² lbs*s	57.8288 ft ² lbs*s
Time	7.73 seconds	7.74 seconds	7.71 seconds	7.80 seconds	7.77 seconds
Angle Traveled (degrees)	92.0°	92.0°	92.0°	92.0°	92.0°
Angle Traveled (radians)	1.61 rad	1.61 rad	1.61 rad	1.61 rad	1.61 rad
Arc Length Traveled by Outer Wheel	57.40 inches	57.40 inches	57.40 inches	57.40 inches	57.40 inches
Arc Length Traveled by Outer Wheel	4.78 ft	4.78 ft	4.78 ft	4.78 ft	4.78 ft
Angular Displacement of Outer Wheel (radians)	4.79 rad	4.79 rad	4.79 rad	4.79 rad	4.79 rad
Average Torque (energy)	7.01 ft lbs	7.22 ft lbs	7.00 ft lbs	7.48 ft lbs	7.44 ft lbs
Total Work (energy)	33.56 ft lbs	34.54 ft lbs	33.48 ft lbs	35.78 ft lbs	35.61 ft lbs
Work/ft (force)	7.02 lbs	7.22 lbs	7.00 lbs	7.48 lbs	7.44 lbs
Drop Hi/Low Work/ft (force)	7.02 lbs	7.22 lbs			7.44 lbs
Average Work/ft (force)			7.23 lbs		

CONCLUSION:

The above listed material **meets/exceeds** both the straight line and turning propulsion requirements set forth in this test, where the surface tested average work per foot value was less than the average work per foot value verses a hard, smooth surface with a grade of 7.1%

Uncertainty:

We undertake all assignments for our clients on a best effort basis. Our findings and judgments are based on the information to us using the latest test methods available. TSI can only ensure the test results for the specific items tested. Unless otherwise noted in the deviations sections of this report, all tests are performed in compliance with stated test method.

Test Report Approval:


Erle Miles, III, Lab Director Testing Services (TSI) LLC

TSI Accreditation:

Our laboratory is accredited by the US Dept. of Commerce, National Institute of Standards and Technology: ISO/IEC 17025:2005. Our code # is: NVLAP 100108-0. TSI is an Organizational Member of ASTM (American Society for Testing and Materials).



Testing Services (TSI) LLC
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TESTING SERVICES

ON-SITE TEST REPORT

Report # 92885
Lab Test Number: 3413-6054
Report Date: December 21, 2023

ASTM 3351-19: Specified Height Impact Attenuation

www.testingservices.us ▪ (706)226-1400
office@testingservices.us

CLIENT:

Company: ForeverLawn
Address: 8007 Beeson Road
Louisville, OH 44641

Test Material:

Turf Identification: Ultra
Infill System: 3.75 lbs/ft² Envirofill
Underlayment System: 3.0" SF Pro Pad
Date Received: 11/7/2023
Sub-base: 3" Aggregate (2" #57 Stone, 1" Fines)

Testing Methods Requested:

Standard: ASTM F3351-19
Standard: ASTM F355-16e1
Testing Services, LLC was instructed by the client to test for the following:
Test Method: Standard Test Method for Playground Surface Impact Testing in Laboratory at Specified Test Height
Standard Test Method for Impact Attenuation of Playing Surface Systems

Sampling Plan:

Sampling Date: 11/7/2023

- Turf and padding were delivered to Testing Services directly.
- The delivered specimens were acclimated to the appropriate temperatures per the test method prior to testing.

Test Apparatus:

Equipment: Triax Touch E Missile System
Accelerometer: LW370974
Calibration Date: 11/2/2023
Manufacturer: Alpha Automation, Inc.
Manufacturer: Dytran
Recalibration Due Date: 11/2/2025

Deviation from Test Method:

State Reason for any Deviation, Additions to, or Exclusions from Test Method

Client requested deviation of sub-base to aggregate.

Test Summary:

Test Method	Condition	Gmax	HIC	Fall Height
ASTM F3351-19	Ambient, Dry	90	505	10'
	Hot, Dry	82	455	10'
	Frozen	119	858	10'

Full test data is reported on page 2 of this report

Critical Fall Height <200 Gmax, <1000 HIC, All Temperature Ranges: 10'

Precision/Bias:

The results reported herein reflect the performance of the tested playground surface at the time of testing and at the temperature(s) and ambient conditions reported. Performance will vary with temperatures, moisture content, and other factors. The precision of the test method, conducted by interlaboratory testing, is estimated to be $\pm 5\%$ for gmax and $\pm 10\%$ for HIC. Users of this test report should be aware of this fact when establishing critical fall height. Testing Services LLC undertakes all assignments for our clients on a best effort basis. Our results, findings, and judgements are based on the information using the latest test methods available. Testing Services LLC can only ensure the test results for the specific items evaluated. Unless otherwise noted in the deviations sections of this report, all tests performed were in compliance with the stated test method.

Decision Rule:

No consideration of uncertainty was taken in our statement of compliance.

Test Report Approval:


Erle Miles, III, Lab Director Testing Services (TSI) LLC

TSI Accreditation:

Our Laboratory is accredited by NVLAP (Lab Code 100108-0), which is part of NIST and the US Dept of Commerce. This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, and any agency of the U.S. Government. The above testing is under the scope of our NVLAP accreditation.



100108-0

Testing Services (TSI) LLC
817 Showalter Avenue
PO Box 1343
Dalton, GA 30721



CLIENT:

Company: ForeverLawn
Address: 8007 Beeson Road
Louisville, OH 44641

Test Material:

Turf Identification: Ultra
Infill System: 3.75 lbs/ft² Envirofill
Underlayment System: 3.0" SF Pro Pad
Date Received: 11/7/2023
Sub-base: 3" Aggregate (2" #57 Stone, 1" Fines)

Test Data: **Average consists of drops 2 & 3; drop 1 is for conditioning only**

Ambient Air Temperature: 71°F Reference Surface Temperature: 72°F ± 5°F Test Date: 12/7/2023

Drop	Velocity	Angle	Drop Height	Gmax	HIC
1	25.5	1	10'	85	469
2	25.5	7	10'	88	494
3	25.5	10	10'	91	516
Average (drops 2&3)				90	505
Surface Temperature: (@ 1 st Impact) 71° (After 3 rd Impact) 71°					

Hot Air Temperature: 120°F Reference Surface Temperature: 120°F ± 5°F Test Date: 12/15/2023

Drop	Velocity	Angle	Drop Height	Gmax	HIC
1	25.5	4	10'	77	411
2	25.5	1	10'	81	448
3	25.3	7	10'	82	461
Average (drops 2&3)				82	455
Surface Temperature: (@ 1 st Impact) 120° (After 3 rd Impact) 120°					

Cold Air Temperature: 28°F Reference Surface Temperature: 25°F ± 5°F Test Date: 12/8/2023

Drop	Velocity	Angle	Drop Height	Gmax	HIC
1	25.5	1	10'	120	868
2	25.5	5	10'	121	867
3	25.5	5	10'	117	848
Average (drops 2&3)				119	858
Surface Temperature: (@ 1 st Impact) 28° (After 3 rd Impact) 28°					

END OF REPORT

Testing Services (TSI) LLC
817 Showalter Avenue
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